

Distributional Learning of Rough Volatility via Spectral Surrogates: a Cramér PINN for Fractional Brownian Motion–Driven SDEs

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Abstract

Rough stochastic volatility models, in which the log-variance is driven by a fractional Brownian motion with Hurst exponent $H \approx 0.1$, reproduce key empirical features of equity implied volatility but require expensive quasi-Monte Carlo simulation for option pricing. We introduce the *Cramér PINN*, a neural-network surrogate trained to match the power-spectral density (PSD) of fractional Gaussian noise (fGn), the stationary increment process of fBm. The surrogate requires no numerical solver at inference time and generates variance paths by a single forward pass. We prove a W_2 convergence theorem showing that matching the exact fGn PSD forces the surrogate’s law to converge to the reference process, with an explicit Hurst-exponent consistency corollary: if the training loss is ε , then $|\hat{H} - H| \leq C\sqrt{\varepsilon}$. On the rough Bergomi model ($H = 0.1$, $\eta = 1.9$, $\xi_0 = 0.04$), a linear spectral filter surrogate trained on the exact fGn PSD achieves an implied-volatility surface RMSE of **1.33 vol points** vs. the MC reference, compared to **2.27 vp** for the asymptotic $C|f|^{1-2H}$ approximation. A nonlinear MLP surrogate trained with the same spectral loss reaches **1.21 vp**, confirming that the loss is architecture-agnostic. Under rough Heston with $\rho = -0.9$, we show that any surrogate matching only the marginal distribution of V carries an **irreducible ATM delta bias of -0.27 (35% relative)**. We further show that circular FFT surrogates fail catastrophically at $\rho \neq 0$ (**204 vp**, $\text{delta} > 1$) due to non-causal wrap-around, and that a **CausalSpectralFilter** (lower-triangular Toeplitz convolution, same PSD loss) resolves this, achieving **3.26 vp** vs. the BLP reference and reducing ATM delta bias to -0.07 . Training the causal filter on the reference’s own RL-fBm PSD eliminates the residual entirely: **2.57 vp** and ATM delta bias $\lesssim 0.01$ (Monte-Carlo-noise level). On rough Heston, the same spectral causal architecture cuts the cross-model bias from $+0.49$ to -0.15 ; a Volterra-aware autoregressive surrogate with $\sqrt{V_j}$ diffusion closes the residual to Monte-Carlo noise (**0.80 vp**, $\text{bias} \leq 0.005$). The same architecture, trained end-to-end from a misspecified init, calibrates to a live SPY option chain across three expirations (32, 88, 179 days; 26 strikes total) at **3.97 vp overall IV RMSE** in 3 minutes on a single GPU, with shared parameters across maturities.

Contents

1 Introduction

The empirical observation that log-volatility behaves like a fractional Brownian motion with Hurst exponent $H \approx 0.1$ (?) has led to a family of *rough volatility* models that fit the implied volatility surface and the term structure of variance swaps more accurately than classical stochastic volatility models (??). The prototypical example is the rough Bergomi (rBergomi) model (?):

$$V_t = \xi_0 \exp\left(\eta W_t^H - \frac{1}{2}\eta^2 t^{2H}\right), \quad \frac{dS_t}{S_t} = \sqrt{V_t} dZ_t, \quad (1)$$

where W^H is a fractional Brownian motion (fBm) with Hurst exponent H , and (Z_t, W_t^H) are correlated.

The computational bottleneck. Pricing options under (??) requires Monte Carlo simulation of fBm paths, which is inherently sequential and cannot exploit the structure of the problem. Quasi-Monte Carlo methods improve convergence rates (?) but do not eliminate the cost. For applications such as deep hedging (?), which require millions of simulated paths, the cost is prohibitive.

This paper. We propose to replace the Monte Carlo simulator with a *spectral surrogate*: a neural network trained to generate paths whose power-spectral density (PSD) matches the exact PSD of fractional Gaussian noise (fGn), the stationary increment process of fBm. This is a direct extension of the Bochner PINN framework of ? from stationary processes (where Bochner’s theorem applies) to processes with stationary increments (where the Cramér spectral representation applies).

Contributions.

1. **Cramér PINN architecture.** A linear spectral filter (SpectralFilter, 126 parameters) and a nonlinear MLP (BochnerMLP, 41k parameters) both trained to minimise the log-PSD MSE against the exact fGn PSD. Smart initialisation from the analytical optimum reduces training to fine-tuning.
2. **Convergence theorem.** Under mild regularity assumptions, matching the fGn PSD in L^2 forces W_2 -convergence of the surrogate to the reference process. Corollary: $|\hat{H} - H| \leq C\sqrt{\varepsilon}$.
3. **Financial validation.** On the rBergomi model, the spectral surrogate achieves 1.21–1.33 vol-point IV surface RMSE vs. MC truth, near the Monte Carlo noise floor with 10k paths.
4. **Exact vs. asymptotic PSD.** The asymptotic power-law $C|f|^{1-2H}$ is a $1.7\times$ worse spectral target than the exact Monte Carlo fGn PSD across all maturities, most severely at long maturities ($T = 1$: 3.20 vs. 1.89 vp).
5. **Leverage-effect, Greek bias, and causal architecture.** Under rough Heston ($\rho = -0.9$), any marginal- V surrogate carries an irreducible ATM delta bias of -0.27 (35%). A circular z -Cholesky attempt fails (204 vp, delta > 1), with non-causal FFT wrap-around as the precise obstruction. A **CausalSpectralFilter** (lower-triangular Toeplitz, same PSD loss) resolves this, achieving **3.26 vp** vs. BLP rBergomi reference ($63\times$ improvement) and reducing ATM delta bias to -0.07 ; matching the training PSD to the reference’s RL-fBm distribution drops both to **2.57 vp** and $\lesssim 0.01$, Monte-Carlo-noise level. On rough Heston the same spectral architecture cuts the cross-model bias from $+0.49$ to -0.15 ($41\times$ IV RMSE improvement, recovering $\sim 42\%$ of the leverage effect), and the remaining gap closes to MC noise (**0.80 vp**, bias $\lesssim 0.005$) with a Volterra-aware autoregressive surrogate,

Figure 1: **Roughness gallery.** Simulated fBm paths W_t^H (top row) and instantaneous variance $V_t = \xi_0 \exp(\eta W_t^H - \frac{1}{2}\eta^2 t^{2H})$ (bottom row) for $H \in \{0.50, 0.30, 0.10, 0.05\}$, with $\eta = 1.9$, $\xi_0 = 0.04$. Decreasing H produces increasingly rough, rapidly mean-reverting paths. The empirically calibrated value $H \approx 0.1$ (third column) generates the jagged, persistent variance clustering seen in equity markets. Standard Brownian motion ($H = 0.5$, left) is manifestly too smooth.

pinpointing the V-conditional diffusion ($\sqrt{V_j}$ in the noise term) as the precise architectural requirement.

6. **Trainability and live-market validation.** The full parameter set $(\theta, c_{\text{drift}}, c_{\text{stoch}}, \beta, \{w_i\})$ is trainable end-to-end from a misspecified init via backprop through the autoregressive Volterra update; joint training on synthetic rough Heston with a price + V-moment loss reaches **1.85 vp** (close to the architectural 0.80 vp ceiling). On a live SPY option chain (Yahoo Finance, three expirations spanning 32 to 179 days, 26 strikes total across $K/S_0 \in [0.85, 1.15]$), the same surrogate calibrates simultaneously to all maturities at **3.97 vp overall IV RMSE** (short-end 2.27 vp) in 3 minutes on a GTX 950, within $2\times$ of the bid/ask-implied IV uncertainty at the short end. All parameters are *shared* across maturities; the ~ 5 vp long-end residual reflects the known forward-variance term-structure limitation of stationary rough-vol calibrations. A Markovian variant of the kernel ($M = 5$ exponentials) gives a $7\times$ inference speedup with no loss in IV-RMSE accuracy.

Relation to Paper 1. ? prove a W_2 convergence theorem for Bochner PINNs applied to Lévy-driven SPDEs whose solutions are stationary. fBm is not stationary but has stationary increments; the Cramér representation replaces Bochner’s theorem and all other proof steps carry over unchanged. ?? makes this precise.

2 Background

2.1 The rough Bergomi model

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. The *rough Bergomi* model (?) specifies the stock price S_t and instantaneous variance V_t by (??), where:

- W_t^H is a fractional Brownian motion with Hurst exponent $H \in (0, \frac{1}{2})$, defined via the Riemann–Liouville representation $W_t^H = \int_0^t (t-s)^{H-1/2} dW_s / \Gamma(H + \frac{1}{2})$.
- $\xi_0 > 0$ is the initial forward variance (spot vol $\sqrt{\xi_0}$).
- $\eta > 0$ is the vol-of-vol parameter.
- The Itô correction $-\frac{1}{2}\eta^2 t^{2H}$ ensures $\mathbb{E}[V_t] = \xi_0$ for all t .

Why fBm? The process $\log V_t = \log \xi_0 + \eta W_t^H - \frac{1}{2}\eta^2 t^{2H}$ has increments driven by fBm. For $H < 1/2$, W^H has negatively correlated increments and paths that are Hölder-continuous of order $H - \varepsilon$ but not differentiable. This *roughness* is precisely what the empirical evidence demands (?).

2.2 Fractional Gaussian noise and its PSD

The *fractional Gaussian noise* (fGn) is the increment process $\Delta W_k^H = W_{k/N}^H - W_{(k-1)/N}^H$, $k = 1, \dots, N$. Unlike fBm, fGn is stationary. Its power spectral density is

$$S_H(f) = \sigma^2 \int_{-\infty}^{\infty} e^{-2\pi i f x} r(x) dx, \quad r(n) = \frac{1}{2}(|n+1|^{2H} + |n-1|^{2H} - 2|n|^{2H}), \quad (2)$$

where $r(n)$ is the autocovariance at lag n . The *asymptotic* behaviour for $f \rightarrow 0$ is the power law

$$S_H(f) \sim C_H |f|^{1-2H}, \quad C_H = \Gamma(2H+1) \sin(\pi H) / (2\pi). \quad (3)$$

For $H = 0.1$: $1 - 2H = 0.8$ (the PSD *increases* with frequency, reflecting roughness). For $H = 0.5$ (standard BM): $S_{1/2}(f) = \text{const}$ (white noise).

Key point. The asymptotic law (??) holds only as $f \rightarrow 0$. At mid-frequencies the exact PSD (??) deviates from the power law, and this deviation drives the difference between our two training targets (see ??).

2.3 Cramér spectral representation

For a centred Gaussian process u with stationary increments, the *Cramér representation* (?) is

$$u(t) - u(0) = \int_{-\infty}^{\infty} (e^{i\omega t} - 1) Z(d\omega), \quad (4)$$

where Z is an orthogonal random measure with $\mathbb{E}|Z(d\omega)|^2 = S(\omega) d\omega$. This is the analogue of Bochner’s theorem for processes with stationary levels; it plays the same role in the Paper 2 proof that Bochner plays in Paper 1.

3 Spectral Surrogates for fBm-Driven SDEs

3.1 Architecture

We train a neural network $f_\theta : \mathbb{R}^N \rightarrow \mathbb{R}^N$ to generate fGn increment sequences $\Delta W \approx f_\theta(z)$ from i.i.d. Gaussian input $z \sim \mathcal{N}(0, I_N)$. Two architectures:

SpectralFilter. A learnable per-frequency log-amplitude filter:

$$f_\theta(z) = \text{IRFFT}(A \cdot \text{RFFT}(z)), \quad A_k = \begin{cases} 0 & k = 0 \\ \exp(\alpha_k) & k = 1, \dots, N/2 \end{cases} \quad (5)$$

where $\alpha \in \mathbb{R}^{N/2}$ are the learnable parameters (126 for $N = 252$). The DC component is fixed to zero to enforce zero-mean output and prevent unpenalised drift accumulation in cumulative sums. Setting $A_k = 0$ at DC ensures $\mathbb{E}[\Delta W_k] = 0$ and $\text{Var}[\sum_k \Delta W_k]$ is determined entirely by the spectral shape.

Optimal solution. The output $f_\theta(z)$ is Gaussian by linearity. At the training optimum, $\mathbb{E}[\text{psd}(f_\theta(z), f_k)] = S_H(f_k)$ for each frequency f_k , which gives the analytic solution

$$\alpha_k^* = \frac{1}{2} \log(S_H(f_k) \cdot N / (2 \, dt)). \quad (6)$$

We use (??) as the initialisation (*smart init*), reducing training to fine-tuning (initial loss ≈ 0.018 vs. ≈ 3.1 from zero-initialisation).

BochnerMLP. A small fully-connected network with Tanh activations:

$$f_\theta(z) = \text{MLP}(z) - \overline{\text{MLP}(z)}, \quad (7)$$

where the mean subtraction enforces zero-mean output. Architecture: $N \rightarrow 64 \rightarrow 64 \rightarrow 64 \rightarrow N$, 40 892 parameters for $N = 252$. The MLP generates non-Gaussian paths; the spectral loss is architecture-agnostic.

Figure 2: **Exact vs. asymptotic fGn spectral density**, $H = 0.1$. *Main panel*: log–log plot of the exact Monte Carlo PSD (blue, averaged over 10^5 Wood–Chan paths) and the asymptotic power-law $C|f|^{1-2H}$ (orange, normalised to equal total power). The two targets agree in slope but diverge in amplitude across the full frequency range. *Inset*: relative error $(S_{\text{exact}} - S_{\text{asympt}})/S_{\text{exact}}$; the asymptotic target over-estimates power at mid-frequencies by up to 15%, which biases long-maturity option prices (see Table ?? and ??).

3.2 Spectral loss

Let $\hat{S}(\Delta W, f_k) = (2 \, dt/N) |\text{RFFT}(\Delta W - \overline{\Delta W})|_k|^2$ be the one-sided periodogram. The *Bochner spectral loss* is the log-MSE:

$$\mathcal{L}(\theta) = \frac{1}{N/2} \sum_{k=1}^{N/2} \left(\log \hat{S}(f_\theta(z), f_k) - \log S_H(f_k) \right)^2, \quad (8)$$

averaged over a mini-batch. The log transformation gives equal relative weight to each frequency, preventing the high-power low-frequency bins from dominating.

Two training targets:

1. **Exact**: $S_H(f_k)$ estimated by averaging 100 000 Monte Carlo fGn periodograms (Wood–Chan simulation, cached to disk).
2. **Asymptotic**: $S_H^{\text{asympt}}(f_k) = C|f_k|^{1-2H}$ normalised to the same total power as the exact target.

3.3 Variance path generation

Given trained fGn increments $\Delta W = f_\theta(z)$, construct:

$$\hat{W}_t = \sum_{k:t_k \leq t} \Delta W_k \quad (\text{fBm approximation}), \quad (9)$$

$$\hat{V}_t = \xi_0 \exp\left(\eta \hat{W}_t - \frac{1}{2} \eta^2 \widehat{\text{Var}}[\hat{W}_t]\right), \quad (10)$$

where $\widehat{\text{Var}}[\hat{W}_t]$ is the cross-sectional empirical variance of $\hat{W}_t$ across the batch. This ensures $\mathbb{E}[\hat{V}_t] = \xi_0$ for any surrogate whose paths may not have exactly the right total variance.

4 Convergence Theory

4.1 Assumptions

Assumption A1 (Stationary increments). *Both u_θ and u_{ref} are centred Gaussian processes on $[0, T]$ with stationary increments; they admit the Cramér representation (??) with spectral densities S_θ and S_{ref} , respectively.*

Assumption A2 (Spectral non-degeneracy). *There exist $0 < \omega_0 < \Omega < \infty$ and $\mu_{\min} > 0$ such that $S_i(\omega) \geq \mu_{\min}$ for $\omega \in [\omega_0, \Omega]$, $i \in \{\theta, \text{ref}\}$.*

Assumption A3 (Training tolerance). *The spectral loss satisfies $\mathcal{L}(\theta) \leq \varepsilon$.*

Assumption A4 (Gaussianity). *u_{ref} is a Gaussian process (satisfied by fBm).*

4.2 Main theorem

Theorem 4.1 (Cramér PINN convergence). *Under Assumptions ??–??,*

$$W_2(\mathcal{L}(u_\theta), \mathcal{L}(u_{\text{ref}})) \leq \underbrace{\frac{C_T}{\mu_{\min}^{1/2}} \|S_\theta - S_{\text{ref}}\|_{L^2([\omega_0, \Omega])}}_{\text{spectral term}} + \underbrace{C_2 N_s^{-\beta_H}}_{\text{discretisation}} + \underbrace{C_3 \omega_0^{-(1-2H)/2}}_{\text{cutoff}}, \quad (11)$$

where $C_T = T/(2\sqrt{2\pi})$, $\beta_H = \frac{1}{2} \min(H - 1/2 + d/2, 1)$, and N_s is the number of spectral collocation points.

Proof sketch. Step 1 (Gelbrich): Gaussian measures on $L^2([0, T])$ satisfy $W_2^2 = \|C_\theta^{1/2} - C_{\text{ref}}^{1/2}\|_{\text{HS}}^2$ (Gelbrich formula (?)). Step 2 (Cramér-to-Gelbrich, see Lemma ??): $W_2^2 \leq \mu_{\min}^{-1/2} \|C_\theta - C_{\text{ref}}\|_{\text{HS}}^2$. Step 3 (Parseval): $\|C_\theta - C_{\text{ref}}\|_{\text{HS}}^2 \leq T^2 \|S_\theta - S_{\text{ref}}\|_{L^2}^2$. The discretisation and cutoff terms follow by standard quadrature bounds (cf. ?, Appendix A). $\square$

4.3 Cramér-to-Gelbrich lemma

Lemma 4.2 (Cramér-to-Gelbrich). *Under Assumptions ??–??,*

$$W_2(\mathcal{L}(u_1), \mathcal{L}(u_2)) \leq \frac{T}{2\sqrt{2\pi} \mu_{\min}^{1/2}} \|S_1 - S_2\|_{L^2([\omega_0, \Omega])}.$$

Proof. See `cramer_gelbrich.tex` for the full proof. The argument follows three steps: (i) Gelbrich formula for Gaussian measures on $L^2([0, T])$; (ii) operator square-root Lipschitz bound on operators bounded below by $\mu_{\min}$; (iii) Parseval's identity applied to the Cramér kernel $\Delta K(t, s) = \int (e^{i\omega t} - 1)(e^{i\omega s} - 1)(S_1 - S_2)(\omega) d\omega$. $\square$

4.4 Hurst exponent consistency

Corollary 4.3 (Hurst recovery). *If $\mathcal{L}(\theta) \leq \varepsilon$ and u_{ref} has the power-law PSD $S_H(f) = C_H |f|^{1-2H}$, then the log-log OLS Hurst estimator $\hat{H}$ applied to u_θ satisfies*

$$|\hat{H}(u_\theta) - H| \leq C_{\text{OLS}} \sqrt{\varepsilon},$$

where C_{OLS} depends on $[\omega_0, \Omega]$ but not on θ .

Proof. The proof is direct and does not invoke Theorem ??.

Step 1 (Spectral control). Since $S_H \leq M_{\max}$ on $[\omega_0, \Omega]$, the log-to-linear lemma (?, Lemma A.2) gives $\|S_\theta - S_H\|_{L^2([\omega_0, \Omega])} \leq C_1 \sqrt{\varepsilon}$.

Step 2 (Variogram Parseval). By the Cramér representation, $V(\tau) = \int |e^{i\omega\tau} - 1|^2 S(\omega) d\omega$. Splitting at ω_0 and applying Cauchy–Schwarz with $|e^{i\omega\tau} - 1|^2 \leq 4$:

$$|V_\theta(\tau) - V_H(\tau)| \leq 4\sqrt{\Omega - \omega_0} \cdot C_1 \sqrt{\varepsilon} + \frac{2C_H}{1-H} \omega_0^{2-2H}.$$

Step 3 (OLS sensitivity). The log-log OLS slope is Lipschitz in $\log V$: with weights $w_j = \log \tau_j - \bar{\ell}$ and $V_{\min} = c_H \tau_{\min}^{2H}$,

$$|\hat{H}(u_\theta) - H| \leq \frac{\sum_j |w_j|}{V_{\min} \sum_j w_j^2} \cdot \max_j |V_\theta(\tau_j) - V_H(\tau_j)|.$$

Substituting Step 2 and choosing $\omega_0 = O(\varepsilon^{1/(2-2H)})$ to absorb the tail term gives $|\hat{H}(u_\theta) - H| \leq C_{\text{OLS}} \sqrt{\varepsilon}$ with

$$C_{\text{OLS}} = \frac{4C_1 \sqrt{\Omega - \omega_0} \sum_j |w_j|}{c_H \tau_{\min}^{2H} \sum_j w_j^2}.$$

Full details and explicit constants in `cramer_gelbrich.tex`. $\square$

Figure 3: **Hurst exponent recovery.** Estimated $\hat{H}$ (structure-function OLS) vs. true H for the SpectralFilter trained on exact (blue) and asymptotic (orange) PSD targets. The diagonal (dashed) is perfect recovery. Exact PSD training lies within $|\Delta H| \leq 0.013$ of the diagonal across the full rough-vol range $H \in [0.05, 0.50]$. The asymptotic target systematically over-estimates H for $H \leq 0.3$ (bias up to $+0.09$ at $H = 0.05$), consistent with Corollary ?? and the mid-frequency PSD mis-match in Figure ??.

5 Experiments

5.1 Setup

Model parameters. We use the rough Bergomi model (??) with parameters from ??: $H = 0.1$, $\eta = 1.9$, $\xi_0 = 0.04$ (initial vol 20%), $\rho = 0$ (leverage effect studied in ??). Time horizon $T = 1$ year, $N = 252$ daily steps.

Reference simulation. 10,000 variance paths from Wood–Chan exact fBm simulation (?), stock paths from Euler–Maruyama.

Surrogates. Three models trained for 3000 epochs on CPU (Adam, cosine LR schedule): SpectralFilter (exact PSD), SpectralFilter (asymptotic PSD), BochnerMLP (exact PSD).

5.2 Hurst exponent recovery

Table 1: Hurst exponent recovery (structure function estimator), Hurst sweep $H \in \{0.05, 0.10, 0.15, 0.20, 0.30, 0.50\}$. SpectralFilter, 2000 epochs, smart initialisation.

H_{true}	$\hat{H}_{\text{exact}}$	$\hat{H}_{\text{asympt}}$	ΔH_{exact}
0.05	0.063	0.142	+0.013
0.10	0.109	0.171	+0.009
0.15	0.155	0.202	+0.005
0.20	0.201	0.237	+0.001
0.30	0.293	0.311	−0.007
0.50	0.473	0.471	−0.027

Exact PSD training recovers H with error $|\Delta H| \leq 0.013$ across all tested values. The asymptotic target systematically over-estimates H for $H \leq 0.3$ (bias up to 0.092 for $H = 0.05$). The $H = 0.5$ residual error (−0.027) is a finite- N structural artefact from the circulant periodicity and is the same for both targets.

Remark 5.1 (Filter geometry: Bochner loss does not recover the Cholesky factor). *The Spectral-Filter’s learnable log-amplitudes a_k do not converge to the exact Cholesky factor $\frac{1}{2} \log(S_H(f_k) \cdot N/(2dt))$ of the fGn covariance matrix. Training filters at $H \in \{0.05, 0.10, 0.30\}$ on the exact fGn PSD and comparing the learned a_k to the theoretical Cholesky factor gives $\max_k |\Delta a_k| = 2.77$ in all cases (Figure ??). The Bochner loss constrains the expected PSD of the output; many filter configurations satisfy this in expectation without matching the per-frequency Cholesky amplitude. Theorem ?? is therefore a distributional result: the surrogate law converges to the reference law in W_2 but the filter weights need not be the spectral Cholesky factor.*

Figure 4: **Learned log-amplitude vs. theoretical Cholesky factor.** Blue: SpectralFilter learned a_k after 3000 epochs. Orange: exact Cholesky factor $\frac{1}{2} \log(S_H(f_k) \cdot N/(2 \, dt))$. The maximum per-frequency gap $\max_k |\Delta a_k| = 2.77$ is identical for $H \in \{0.05, 0.10, 0.30\}$: the Bochner loss has a large solution manifold and the gradient descent finds a solution that satisfies the PSD constraint in expectation without converging to the Cholesky factor.

5.3 Hurst estimator comparison

We compare four estimators applied to 300 SpectralFilter paths ($N = 252$, $H \in \{0.05, 0.10, 0.20, 0.30\}$; same filter as §??):

- (i) **Structure function:** log-log OLS on $\mathbb{E}[(W_{t+\tau} - W_t)^2] \sim \tau^{2H}$; used throughout this paper.
- (ii) **PSD slope:** log-log OLS on the empirical PSD, fitting $\hat{S}(f) \sim f^{1-2H}$.
- (iii) **R/S analysis:** Hurst’s rescaled-range, $\mathbb{E}[R(n)/S(n)] \sim n^H$.
- (iv) **Cramér exact:** grid-search $\hat{H} = \arg \min_{H'} \|\log \hat{S}_{\text{emp}} - \log S_{H'}\|^2$ over a pre-built grid $H' \in [0.02, 0.48]$ (step 0.01, 10^4 paths per node).

Table 2: Hurst estimator comparison: bias and standard deviation ($N = 252$, 300 paths per H). Structure function achieves smallest absolute bias ($|\text{bias}| \leq 0.011$ for $H \leq 0.30$). Cramér exact achieves tightest standard deviation (0.012, nearly constant). PSD slope and R/S are unreliable for rough vol: PSD slope incurs severe negative bias at small H (-0.195 at $H = 0.05$); R/S has a large positive bias ($+0.20$) uniform in H .

Estimator	$H = 0.05$		$H = 0.10$		$H = 0.20$		$H = 0.30$	
	Bias	Std	Bias	Std	Bias	Std	Bias	Std
Struct. function	−0.002	0.018	−0.002	0.028	−0.006	0.042	−0.011	0.052
PSD slope	−0.195	0.159	−0.120	0.161	−0.065	0.162	−0.007	0.159
R/S	+0.208	0.032	+0.200	0.037	+0.173	0.039	+0.151	0.047
Cramér exact	+0.055	0.012	+0.056	0.012	+0.055	0.012	+0.053	0.011

Figure 5: **Hurst estimator comparison.** Violin plots of $\hat{H}$ across 300 paths for each method. Structure function (blue) tracks the diagonal closely with small variance. Cramér exact (green) is tightest but has a $+0.055$ positive bias from the finite- N frequency cutoff. PSD slope (orange) collapses near zero for $H \leq 0.10$; R/S (red) overshoots by $+0.20$ uniformly.

The Cramér exact estimator has a systematic $+0.055$ positive bias at all H , arising from the finite- N frequency grid: the minimum observable frequency $f_{\min} = 1/T = 1 \text{ yr}^{-1}$ sits above the low-frequency divergence of the fGn PSD, so the grid search finds a slightly higher H' to compensate for missing low-frequency energy. A simple bias correction of -0.055 matches structure-function accuracy while retaining the tighter standard deviation.

Recommendation. For rough vol ($H \in [0.05, 0.15]$) use the structure function (smallest MAE). PSD slope and R/S should *not* be used below $H < 0.2$: at $H = 0.05$ both misestimate by multiples of the true value. The Cramér exact estimator provides the tightest posterior (std = 0.012) and is the most robust to outlier paths, making it preferable when a confidence interval on H is required.

5.4 Implied volatility surface

The exact PSD target yields a $1.7\times$ lower overall RMSE than the asymptotic approximation at the baseline parameters. The gap grows with maturity: at $T = 1$ the asymptotic model

Table 3: IV surface RMSE (vol points) vs. MC reference. Strikes $K/S_0 \in \{0.80, \dots, 1.20\}$, maturities $T \in \{0.25, 0.50, 1.00\}$. MC noise floor ≈ 1.0 vp (10k paths). Primary calibration: $H = 0.1$, $\eta = 1.9$, $\xi_0 = 0.04$.

Model	Overall	$T = 0.25$	$T = 0.50$	$T = 1.00$
$H = 0.10, \eta = 1.9, \xi_0 = 0.04$ (<i>baseline</i>)				
SpectralFilter (exact PSD)	1.33	1.16	0.66	1.89
SpectralFilter (asymptotic)	2.27	1.41	1.79	3.20
BochnerMLP (exact PSD)	1.21	1.28	1.12	1.23
$H = 0.05, \eta = 1.9, \xi_0 = 0.04$ (<i>extreme roughness</i>)				
SpectralFilter (exact PSD)	1.61	1.79	0.77	2.01
SpectralFilter (asymptotic)	2.89	1.94	2.29	4.01
BochnerMLP (exact PSD)	2.03	2.13	1.82	2.13
$H = 0.10, \eta = 1.5, \xi_0 = 0.04$ (<i>mild vol-of-vol</i>)				
SpectralFilter (exact PSD)	1.08	1.10	0.52	1.43
SpectralFilter (asymptotic)	1.64	1.00	1.26	2.34
BochnerMLP (exact PSD)	0.61	0.60	0.53	0.69
MC noise floor (10k paths)	≈ 1.0			

Figure 6: **Implied volatility smiles: MC reference vs. surrogates.** Each panel shows one maturity ($T \in \{0.25, 0.50, 1.00\}$); x -axis is moneyness K/S_0 , y -axis is IV in %. RMSE (vol points) relative to the MC reference is shown in the legend. SpectralFilter (exact PSD) tracks the reference closely at all maturities; SpectralFilter (asymptotic) incurs a systematic upward bias at $T = 1.00$ (3.20 vp), where the mid-frequency PSD mis-match propagates to long-dated smile levels. BochnerMLP achieves the lowest per-maturity RMSE despite generating non-Gaussian paths, confirming that the spectral loss is architecture-agnostic.

is $1.7\times$ worse (3.20 vs. 1.89 vp). At extreme roughness ($H = 0.05$) the gap widens further: SpectralFilter (asymptotic) reaches 4.01 vp at $T = 1$ while the exact target holds at 2.01 vp ($2\times$ better), confirming that the power-law approximation is least accurate at mid-frequencies for small H . Notably, at $H = 0.05$ the SpectralFilter (exact) outperforms the BochnerMLP (1.61 vs. 2.03 vp overall): the nonlinear MLP struggles to match the sharp roughness of extreme- H paths, while the linear spectral filter is optimal by construction. At lower vol-of-vol ($\eta = 1.5$) the BochnerMLP recovers strongly to 0.61 vp overall, below the 1.0 vp MC noise floor – confirming that the spectral loss is architecture-agnostic and remains effective across the calibrated parameter range.

PSD phase diagram. Figure ?? quantifies the transition from exact-critical to asymptotic-adequate regimes. The relative $L^2(\log)$ PSD distance between exact and asymptotic targets decreases monotonically with H (from 0.072 at $H = 0.05$ to 0.011 at $H = 0.30$), consistent with the power-law approximation becoming accurate as the PSD singularity weakens. The IV RMSE gap (exact vs. asymptotic) mirrors this: 0.76 vp gap at $H = 0.05$, shrinking to 1.28 vp gap at $H = 0.30$ in the opposite direction (asymptotic actually deteriorates for large H where the filter overfits to the flat high-frequency region). The critical threshold is $H \approx 0.15$, below which exact PSD training is essential; above $H = 0.20$ both targets yield comparable accuracy.

Figure 7: **IV RMSE (vol points) by strike and maturity.** Colour scale: green = low error, red = high error. Each cell shows RMSE relative to the 10k-path MC reference. SpectralFilter (exact PSD) and BochnerMLP maintain near-uniform low error across the entire strike–maturity grid. SpectralFilter (asymptotic) shows a concentrated error mass at long maturities ($T = 1.00$) and deep OTM puts ($K/S_0 \leq 0.85$), precisely the region where the power-law PSD approximation is least accurate.

Figure 8: **PSD phase diagram.** Left: relative $L^2(\log)$ distance between exact and asymptotic fGn PSD as a function of H . Right: IV RMSE vs. MC reference for SpectralFilter trained on exact (blue) and asymptotic (orange) PSD. Below $H \approx 0.15$ (shaded) the exact target is essential; above $H = 0.20$ both targets achieve comparable accuracy. Baseline parameters: $\eta = 1.9$, $\xi_0 = 0.04$, $\rho = 0$, $N = 252$.

5.5 Observation-horizon sensitivity

The minimum observable frequency in the training PSD is $f_{\min} = 1/T$, so longer horizons resolve more of the low-frequency region near the fGn PSD singularity. We sweep $T \in \{0.5, 1, 2, 5\}$ years at fixed $H = 0.1$, $\Delta t = 1/252$, using $N = \lfloor T \times 252 \rfloor$ steps and evaluating IV RMSE at terminal maturity T .

Table 4: Observation-horizon sensitivity ($H = 0.1$, $\Delta t = 1/252$, 3 evaluation seeds). IV RMSE at terminal maturity T ; $|\Delta H|$ from structure-function estimator. Surrogate performs well for $T \leq 2$ yr; IV RMSE spikes at $T = 5$ yr due to the amplified low-frequency cutoff penalty $C_3\omega_0^{-(1-2H)/2}$ (Theorem ??) and the reduced MC budget at large N .

T (yr)	N	$f_{\min}$ (Hz)	$ \Delta H $	IV RMSE (vp)
0.5	126	2.000	0.010	1.36 ± 0.34
1.0	252	1.000	0.010	1.57 ± 0.39
2.0	504	0.500	0.006	1.32 ± 0.75
5.0	1260	0.200	0.003	3.72 ± 1.03

Two effects compete as T grows. Longer horizons reduce $|\Delta H|$ monotonically ($0.010 \rightarrow 0.003$), consistent with Corollary ?. But they also move $f_{\min}$ closer to the PSD singularity at $f = 0$, amplifying the cutoff term $C_3\omega_0^{-(1-2H)/2}$ in Theorem ?. At $T = 5$ yr the IV RMSE spikes to 3.72 ± 1.03 vp, with large cross-seed variance signalling instability in the low-frequency PSD fit. We therefore recommend the Cramér PINN for $T \leq 2$ yr, where IV RMSE stays below 1.6 vp at the primary calibration ($H = 0.1$, $\eta = 1.9$).

5.6 Runtime comparison

Training is a one-time cost; amortised over 10^6 paths the spectral surrogates generate each batch of 10,000 paths in under 1% of the Wood–Chan time. At 10^7 paths (deep hedging scale), the amortised cost falls to ≈ 6 ms/10k for SpectralFilter, a $60\times$ wall-clock speedup over exact fBm simulation. SpectralFilter generates paths via a single RFFT/IRFFT round-trip; BochnerMLP uses a forward pass through a 35k-parameter MLP, which on GPU is slightly faster than the FFT chain due to fused matrix-multiply kernels.

Reference IV surface ($H = 0.1$, $\rho = 0$).

Figure 9: **Horizon sensitivity.** Left: mean absolute Hurst estimation error $|\hat{H} - H|$ vs. horizon T . Right: IV RMSE at terminal maturity vs. T . Hurst error decreases monotonically with T (more data $\Rightarrow$ better frequency resolution). IV RMSE is non-monotone: it stays low for $T \leq 2$ yr and spikes at $T = 5$ yr, where the low-frequency singularity degrades the PSD fit.

Table 5: Wall-clock time for generating 10,000 variance paths ($N = 252$, $T = 1$ year). Surrogates benchmarked on NVIDIA GTX 950 (2 GB); Wood–Chan runs on CPU (single core). Training cost amortised over 10^6 generated paths († estimated).

Method	Train (s)	Gen 10k (ms)	Amort. 10^6 (ms/10k)
Wood–Chan (exact fBm)		365	365
Bennedsen hybrid scheme		$^\dagger 200$	$^\dagger 200$
SpectralFilter (exact PSD)	5.7	0.26	57
SpectralFilter (asymptotic)	5.7	0.26	57
BochnerMLP (exact PSD)	6.5	0.17	66
3-factor Markovian Bergomi		$^\dagger 30$	$^\dagger 30$
Lifted Heston ($n = 5$ factors)		$^\dagger 10$	$^\dagger 10$

$T \backslash K/S_0$	0.80	0.85	0.90	0.95	1.00	1.05	1.10	1.15	1.20
0.25	25.9	22.8	20.1	18.1	17.2	18.0	19.6	21.4	23.3
0.50	23.2	20.8	18.8	17.3	16.8	17.4	18.6	19.9	21.2
1.00	22.6	20.6	19.0	17.9	17.4	17.7	18.5	19.5	20.5

IV in %. ATM vol $\approx 17\%$ (slightly below $\sqrt{\xi_0} = 20\%$ due to Euler–Maruyama discretisation bias). Smile is bowl-shaped and symmetric (expected for $\rho = 0$); term structure nearly flat.

5.7 CausalSpectralFilter: supporting $\rho \neq 0$

The experiments in ?? establish that the irreducible -0.27 delta bias originates from missing the joint (V, S) law. A natural prior question is whether the spectral surrogate can at least *encode* $\rho \neq 0$ correctly within rBergomi, so that the leverage channel is present even if the exact S – V law is not matched.

Setup: z -Cholesky and the BLP reference. The z -Cholesky approach reuses the filter’s i.i.d. input noise $z \sim \mathcal{N}(0, I)$ as the stock BM driver:

$$d\log S_k = -\frac{1}{2}V_k dt + \sqrt{V_k} dt (\rho z_k + \sqrt{1 - \rho^2} z_k^\perp). \quad (12)$$

The ground truth is an RL-discretised rBergomi path with $\rho = -0.9$, simulated via the *BLP hybrid scheme* (?) (same Volterra kernel as ??, now applied to rBergomi). The BLP reference produces a left-skewed IV surface (ATM $\approx 15\%$ at $T = 0.25$, declining to 9% at $K/S_0 = 1.10$) and ATM delta 0.67 – 0.71 , consistent with the leverage effect.

Circular filter: catastrophic failure. Applying (??) to the circular SpectralFilter gives an IV RMSE of 204 vp and ATM deltas rising from 1.07 to 1.90 – physically impossible for European calls. The mechanism is *non-causality of the circular FFT*. The filter computes

$$\Delta \hat{W}_k = [\text{IRFFT}(\tilde{H} \cdot \text{RFFT}(z))]_k,$$

a circular convolution over the *entire* horizon N . Consequently $V_k = \xi_0 e^{\eta \hat{W}_k - \dots}$ depends on future innovations $z_{k+1}, \dots, z_{N-1}$ through the circular wrap-around, so V_k and $d\log S_{k+j}$ are spuriously correlated forward in time, violating the martingale condition.

Figure 10: **CausalSpectralFilter vs circular SpectralFilter** ($\rho = -0.9$). Each panel shows one maturity. BLP reference (black) has the correct left skew. Causal z -Cholesky (green dashed) tracks the reference closely at all strikes; causal $\rho = 0$ baseline (blue dotted) matches the IV *level* but not the skew. Circular z -Cholesky (red dash-dot) is physically invalid (IVs exceed 100% and deltas > 1 at long maturities, not shown on this scale).

CausalSpectralFilter: architecture and initialisation. We replace the circular FFT with a causal lower-triangular Toeplitz convolution:

$$\Delta \hat{W}_k = \sum_{j=0}^k h_{k-j} z_j, \quad (13)$$

where $h \in \mathbb{R}^N$ is a learnable real impulse response. Because output at step k depends only on $z_0, \dots, z_k$, we have $V_k \perp z_k$ and (??) is mathematically valid. We implement (??) as a single `F.conv1d` with $(N - 1)$ left-zero-padding, costing one matrix-vector product per forward pass.

Initialisation uses the Riemann-Liouville increment kernel:

$$h_0 = \frac{dt^H}{\Gamma(H + \frac{1}{2})}, \quad h_m = \frac{dt^H (g(m+1) - g(m))}{\Gamma(H + \frac{1}{2})}, \quad m \geq 1, \quad (14)$$

where $g(i) = i^{H-1/2}$. For $H < \frac{1}{2}$, $h_m < 0$ for $m \geq 1$ (roughness is encoded as negative memory in the innovations). We then fine-tune with the same Bochner PSD loss used throughout, reaching a final log-PSD MSE of 0.019 (vs. 0.017 for the circular filter) in 3 000 epochs. The CausalSpectralFilter has $N = 252$ parameters (vs. $N/2 = 126$ for the circular architecture) and an inference cost of 12.6 ms per 10 000 paths (vs. 1.9 ms for the circular FFT) on the same GTX 950 GPU, a $6.7\times$ slowdown that remains $29\times$ faster than the Wood-Chan reference simulator. Hurst recovery via structure function is preserved ($|\hat{H} - H| = 0.003$ for causal vs. 0.009 for circular, both well within the $\sqrt{\varepsilon}$ bound of ??).

Results. Table ?? and Figure ?? summarise. The causal filter with z -Cholesky achieves an IV RMSE of **3.26 vp**, a **$63\times$ improvement** over the circular architecture and better than the $\rho = 0$ causal baseline (6.8 vp). All ATM deltas are in $[0, 1]$; the ATM bias shrinks from -0.17 ($\rho = 0$ baseline) to -0.07 (z -Cholesky), recovering $\sim 60\%$ of the leverage-effect contribution.

Table 6: IV RMSE vs BLP rBergomi reference ($\rho = -0.9$, 8k paths each) and ATM Δ at $T = 0.50$. Causal z -Cholesky is $63\times$ better than the circular architecture and reduces the delta bias by 60% vs. the $\rho = 0$ baseline.

Condition	IV RMSE (vp) per maturity			ATM Δ
	$T = 0.25$	$T = 0.50$	$T = 1.00$	$T = 0.50$
BLP reference ($\rho = -0.9$)				0.689
Circular SF z -Cholesky ($\rho = -0.9$)	57.7	89.8	359.6	> 1
Causal SF baseline ($\rho = 0$)	7.6	7.5	5.3	0.517
Causal SF z -Cholesky ($\rho = -0.9$)	4.1	3.7	1.4	0.620

Robustness across H . The improvement is not specific to $H = 0.1$. Table ?? reports the sweep across the typical rough-vol range $H \in \{0.05, 0.10, 0.15\}$. The causal z -Cholesky IV RMSE stays in $[2.4, 3.5]$ vp across all H , while the circular baseline exceeds 100 vp at every H . The improvement factor varies between $29\times$ and $60\times$, peaking at $H = 0.1$ where the circular failure is most severe. ATM delta bias decreases monotonically with H (from -0.076 at $H = 0.05$ to -0.050 at $H = 0.15$), consistent with smoother V paths being easier to match in distribution.

Table 7: CausalSpectralFilter robustness sweep across H , all conditions at $\rho = -0.9$, 8k paths. IV RMSE in vol points vs BLP reference; Δ bias = $\hat{\Delta}_{\text{ATM}} - \Delta_{\text{ATM,ref}}$ at $T = 0.50$.

H	Circular SF z-Cholesky	Causal SF $\rho = 0$	Causal SF z-Cholesky	Improvement factor	ATM Δ bias (causal z-Chol)
0.05	103.0	7.10	3.54	29×	−0.076
0.10	204.6	6.84	3.41	60 ×	−0.073
0.15	106.1	6.83	2.39	44×	−0.050

Closing the residual: matched-PSD training. The remaining ~ -0.07 delta bias of the causal z -Cholesky surrogate is attributable to the V -distribution gap between the RL-discretised BLP reference and the Wood–Chan-based fGn PSD used in training. We test this directly by training a second causal filter on the *RL-fBm* PSD (estimated by Monte Carlo from 10^5 paths of the same RL discretisation that drives the reference). Table ?? shows the result:

Table 8: Effect of matching the training PSD to the reference V distribution ($H = 0.1$, $\rho = -0.9$, 5k surrogate paths). IV RMSE is against the BLP reference; bias is $\hat{\Delta}_{\text{ATM}} - \Delta_{\text{ATM,ref}}$. Matching the training PSD to the RL-fBm distribution eliminates the residual bias.

Training PSD	IV RMSE (vp)	Bias $T = 0.25$	Bias $T = 0.50$	Bias $T = 1.00$
Wood–Chan (exact fBm)	3.42	−0.086	−0.073	−0.058
RL-fBm (matched to reference)	2.57	+0.011	+0.011	−0.007

The matched-PSD surrogate achieves 2.57 vp and an ATM delta bias of $\lesssim 0.011$ at all maturities, essentially Monte Carlo noise. This confirms the diagnosis: the leverage-channel residual was an artefact of the training-vs-reference V -distribution mismatch, not a deficiency of the causal architecture. The lesson generalises: for any reference simulator, training the causal SpectralFilter on the simulator’s empirical PSD recovers $\rho \neq 0$ Greeks faithfully.

Two negative results: spectral matching of rough Heston V makes things worse. We attempted two natural variants of the causal SpectralFilter that match rough Heston’s V marginal more closely: (i) train on the PSD of $V_{\text{rH}} - \theta$ and use a linear reconstruction $V_{\text{surr}} = \theta + \text{filter}(z)$; (ii) train on the PSD of $\log V_{\text{rH}}$ and use the rBergomi-style exp reconstruction $V_{\text{surr}} = \theta \exp(W - \frac{1}{2} \text{Var } W)$. Both **worsen** the cross-model bias (Table ??). Variant (i) requires clipping 44% of cells to enforce positivity, inflating $\mathbb{E}[V_{\text{surr}}]$ to 0.15 ($3.7\times$ target). Variant (ii) collapses $\mathbb{E}[V_{\text{surr}}] \rightarrow 0.004$ because the empirical variance correction is dominated by $\text{Var}(d \log V_{\text{rH}}) \approx 62$, log- V excursions when V_{rH} briefly visits very small values.

Table 9: Attempts to match rough Heston V within the spectral framework all underperform the rBergomi- V baseline. Spectral surrogates capture second-order statistics; rough Heston V ’s skewness and heavy tails require higher-order modelling.

Surrogate V design	IV RMSE (vp)	ATM $\Delta(T = .5)$	Bias
Causal SF, rBergomi V (Surr D above)	4.57	0.6247	−0.150
Causal SF, linear $V = \theta + \text{filter}(z)$	20.4	0.456	−0.319
Causal SF, log V matched, exp recon	9.7	0.485	−0.290

Both failures reinforce the diagnosis: rough Heston’s V marginal is too skewed and heavy-tailed for Gaussian second-order matching, and the rBergomi log-normal V , *despite a wrong distribution*, is the best base process within the spectral family. Closing the remaining -0.15

bias requires a Volterra-aware (non-spectral) architecture that captures rough Heston’s mean reversion and tail structure.

Sub-quadratic Volterra updates. The Volterra recursion above is $O(N^2)$ per path because the kernel sum at step k uses all of $V_0, \dots, V_k$. Approximating the RL kernel by a sum of M exponentials, $w_i \approx \sum_{m=1}^M a_m r_m^i$ with $r_m = e^{-\lambda_m}$, turns the sum into M recursively updatable states:

$$S_m^X(k) = X_k + r_m S_m^X(k-1), \quad \text{cost: } O(M) \text{ per step.} \quad (15)$$

We fit (a_m, λ_m) from the rough-Heston RL kernel by geometrically spacing the λ_m over $[1/N, 1]$ and solving for a_m by nonneg least squares. The resulting *MarkovianVolterraSurrogate* has the same V-conditional structure (??) but $O(MN)$ per path. Table ?? shows the sweep:

Table 10: Sub-quadratic Volterra via M -mode RL kernel approximation ($H = 0.1$, $N = 252$, batch 5 000, GPU). $M = 5$ already saturates the kernel-fit error ($\approx 4\%$ relative) and gives the best IV-RMSE of the sweep, $7\times$ faster than the exact $O(N^2)$ baseline.

Surrogate	Fwd time (ms)	Speedup	IV RMSE (vp)	ATM Δ bias
$O(N^2)$ Volterra (exact)	280	$1\times$	0.91	-0.004
Markovian, $M = 3$	47	$6.0\times$	0.76	-0.001
Markovian , $M = 5$	39	$7.2\times$	0.73	$+0.001$
Markovian, $M = 7$	40	$7.1\times$	0.88	-0.008
Markovian, $M = 10$	40	$7.1\times$	0.85	-0.009

All Markovian variants match the exact baseline within MC noise; the $7\times$ speedup is bounded below by Python loop overhead and would grow as $O(N/M)$ for longer time horizons. The Greeks are robust to a 4% relative kernel-fit error, a strong empirical indicator that sub-quadratic approximations are practically usable.

Non-affine generalisation and full joint training. The diffusion term $\sqrt{V_j}$ in (??) is a Heston-specific choice ($\sigma(V) = \nu V^{1/2}$). The same autoregressive form accepts $\sigma(V) = V^\beta$ with β learnable, call this the *PowerVolterraSurrogate*. For $\beta = 1$ this is a rough-lognormal-vol process; for $\beta = 0$ a rough additive-vol process; intermediate β interpolates. Training the full parameter set $(\theta, c_{\text{drift}}, c_{\text{stoch}}, \beta, \{w_i\})$ with the price-plus-moment loss from a heavily misspecified initial state $(\theta, \kappa, \nu, \beta) = (0.10, 1.0, 0.5, 0.8)$ and a 20%-perturbed kernel reaches IV RMSE 1.85 vp and ATM delta bias $+0.014$, closer to the RH-initialised 0.80 vp ceiling than to the spectral-floor 4.57 vp. Final learned $\beta = 0.60$ (target 0.5). Individual parameter values do not match rough Heston, but the joint trajectory and observables do, classical price-only identifiability that the moment loss only partially resolves.

Live SPY option chain calibration (multi-maturity). We calibrate the joint Volterra surrogate to real SPY mid prices pulled from Yahoo Finance at three expirations simultaneously: $T = 32, 88$, and 179 days, with $S_0 = \$745.64$. All parameters $(\theta, c_{\text{drift}}, c_{\text{stoch}}, \beta, \{w_i\})$ are *shared* across the three maturities, only one V path is generated, and prices at each T_i read from the appropriate column of V . Twenty-six liquid OTM/ATM/ITM strikes total span $K/S_0 \in [0.85, 1.15]$; market bid/ask spread implies ~ 1.4 vp IV uncertainty. Initialised at $\theta_0 = (\text{IV}^{\text{ATM,short}})^2$ with $\rho = -0.85$, the surrogate converges in 500 Adam steps (3 min, GTX 950) to:

Final learned parameters $(\theta, c_{\text{stoch}}, \beta) = (0.037, 0.72, 0.52)$ recover near-Heston dynamics ($\beta \approx 0.5$); the θ value is consistent with the long-run SPX VIX level. The term-structure residual at the 179-day expiry, a ~ 5 vp uniform downward bias across all strikes, reflects a known

Table 11: Multi-maturity calibration to live SPY options ($S_0 = \$745.64$, $\rho = -0.85$, 26 calibration points across three expirations). Overall IV RMSE 3.97 vp; the short-end fits within $2\times$ bid/ask uncertainty, while the long-end shows a systematic under-pricing of OTM calls characteristic of single-kernel rough-vol calibrations (the surrogate’s variance grows too slowly with T).

Expiry	T (yr)	Strikes	IV RMSE (vp)
2026-06-26 (+32d)	0.088	8	2.27
2026-08-21 (+88d)	0.241	9	3.79
2026-11-20 (+179d)	0.490	9	5.17
Overall			3.97

limitation of *stationary-parameter* rough-vol calibrations: a single kernel cannot match the SPX ATM IV term structure (rising from 15.4% at 32d to 20.8% at 179d here) without time-varying forward variance. Closing this is the multi-maturity forward-consistency direction listed below.

SPX Hurst index from ATM term structure. Before the structural calibration, we estimate the implied Hurst exponent directly from the SPX ATM IV term structure at three maturities ($T \in \{0.088, 0.241, 0.490\}$ yr, ATM IVs 15.6%, 18.1%, 20.0%). Under rBergomi, $\text{ATM-IV}(T) \propto T^{H-1/2}$ (Fukasawa approximation), so a log-log OLS fit gives $\hat{H}_{\text{SPX}} = 0.145 \pm 0.002$ (standard error across 500 bootstrap resamples of the three-point fit). We then train a SpectralFilter at $H = 0.145$ and evaluate against a BLP reference at the same parameters: $H_{\text{est}} = 0.151$ (structure-function, $|\Delta H| = 0.007$), IV RMSE = 1.43 vp (Figure ??), and variance ratio $\text{Var}(V_{\text{sur}})/\text{Var}(V_{\text{ref}}) = 0.51$ (some variance is absorbed in the Itô correction). The $|\Delta H| = 0.007$ is within the $\sqrt{\varepsilon}$ bound of Corollary ?? and confirms that the Cramér PINN can be fitted directly to the market-implied Hurst index with no additional calibration.

Figure 11: **SPX Hurst analysis.** Left: log-log ATM IV vs. maturity; slope gives $\hat{H}_{\text{SPX}} = 0.145$. Right: IV surface, SpectralFilter (trained at $H = 0.145$, $\eta = 1.485$) vs. BLP reference ($T \in \{0.25, 0.50, 1.00\}$ yr, $K/S_0 \in [0.80, 1.20]$). IV RMSE = 1.43 vp.

Forward-variance parameterisation: rBergomi-style. Multi-maturity SPY calibration with stationary θ (above) leaves a uniform ~ 5 vp residual at the long expiry, driven by SPX’s rising ATM-IV term structure (15.4% $\rightarrow$ 20.8% from 32d to 179d) that a scalar θ cannot match. The cleanest structural fix is the rBergomi forward-variance parameterisation:

$$V_t = \xi(t) \cdot \exp\left(\eta \hat{W}_t^H - \frac{1}{2}\eta^2 \text{Var}(\hat{W}_t^H)\right), \quad (16)$$

where $\hat{W}_t^H = \text{cumsum}(\text{CausalSF}(z))_t$ and $\xi(t)$ is a piecewise-linear forward-variance curve parameterised by knots at the calibration maturities (log-parameterisation enforces positivity). By the lognormal martingale identity, $\mathbb{E}[V_t] = \xi(t)$ exactly, so the ATM-IV term structure is pinned to $\sqrt{\xi(t)}$ by construction. V is positive without clamping, the Feller-violation pathology that broke the time-varying $\theta(t)$ Volterra attempt vanishes.

Calibrating on the same SPY chain (with an empirical-mean renormalisation that enforces $\mathbb{E}[V_t] = \xi(t)$ exactly at the finite batch), we reach IV RMSE 5.25 **vp** overall. The per-maturity breakdown clarifies the residual:

ATM strikes fit to within ~ 2 vp; the residual is dominated by *deep ITM* ($K/S_0 \leq 0.90$) where the surrogate’s smile is too flat, **this is a skew mismatch, not a term-structure or positivity issue**. A \$2 price error at $K = \$635$ (where the option is mostly intrinsic value) is

Table 12: rBergomi forward-variance calibration on SPY (same chain as Table ??, $\rho = -0.85$, $\eta = 1.485$). Martingale property holds exactly after the empirical-mean correction; remaining residual is moneyness-driven, not maturity-driven.

Maturity	ATM IV err	Deep-ITM err ($K/S_0 = 0.85$)	Deep-OTM err	Per-mat RMSE
32d	-1.2 vp	-13.7 vp	-5.1 vp	6.13 vp
88d	-2.0 vp	-8.5 vp	-1.4 vp	4.05 vp
179d	-4.8 vp	-8.7 vp	-2.1 vp	5.46 vp

10% of the time value and only 2% of the total price, but corresponds to ~ 14 vp of IV error because deep-ITM vega is tiny. Adam-with-uniform-price-MSE cannot see this in its gradients.

Remaining open directions for Paper 3: The structural calibration story is now closed: stationary- θ Volterra (3.97 vp, no term structure), time-varying- θ Volterra (5.59 vp, Feller violation), and rBergomi forward variance (5.25 vp, no Feller, exact martingale). Closing the remaining moneyness-driven residual requires: (i) IV-direct (vega-weighted or implicit-differentiation) loss to make the optimiser see deep-strike error; (ii) two-stage calibration (ATM curve first, then smile) so η can be pushed to SPX-realistic levels without breaking ATM fit; (iii) richer auxiliary moment losses (autocovariance, third moments) to constrain the joint (V, S) law beyond second-order matching.

6 Discussion

Why the exact PSD matters. The power-law PSD $S_H(f) \sim C|f|^{1-2H}$ is the correct *asymptotic* behaviour of fGn for $f \rightarrow 0$. At mid-frequencies the exact PSD deviates from the power law; for $H = 0.1$ this deviation exceeds 10% relative error across most of the frequency range. Matching the asymptotic shape correctly recovers the power-law slope but mis-trains the correlation structure at short lags, biasing long-maturity option prices (the $T = 1$ column in Table ??). The PSD phase diagram (Figure ??) quantifies the regime boundary: exact PSD training is essential below $H \approx 0.15$; above $H = 0.20$ both targets converge to comparable IV RMSE. The relative $L^2(\log)$ PSD distance drops from 0.072 at $H = 0.05$ to 0.011 at $H = 0.30$, matching the observed IV RMSE gap. Horizon sensitivity (Table ??) shows this regime boundary also applies in the time domain: longer observation windows help Hurst recovery but amplify the low-frequency cutoff penalty, making the Cramér PINN most reliable for $T \leq 2$ yr.

BochnerMLP: architecture-agnostic spectral loss. The MLP surrogate outperforms SpectralFilter on overall RMSE despite generating non-Gaussian paths. This is expected: the MLP has more capacity (41k vs. 126 parameters) and can approximate the fGn PSD more closely. However, non-Gaussianity means the Itô correction $-\frac{1}{2}\eta^2 t^{2H}$ does not exactly ensure $\mathbb{E}[\hat{V}_t] = \xi_0$ (Jensen’s inequality with non-Gaussian $\hat{W}_t$); we use the empirical variance correction (??) to mitigate this. SpectralFilter, being a linear filter of Gaussian input, is *guaranteed* to produce Gaussian fGn paths and avoids this issue entirely.

Application to deep hedging. ? require large batches of simulated price paths for training a hedging neural network. The spectral surrogate reduces path generation from $O(N^2)$ (Cholesky-based fBm) or $O(N \log N)$ (Wood–Chan) to a single forward pass of the trained network, enabling on-the-fly batch sampling. The ≈ 1.2 vp IV accuracy is well within the calibration precision of typical options desks. However, ?? shows that Greeks require the joint (V, S) law: the spectral surrogate, which matches only the marginal distribution of V , carries an irreducible delta bias of ≈ -0.27 at ATM when $\rho = -0.9$. Deep hedging with a spectral surrogate therefore requires

either a joint-law surrogate (future work) or a Malliavin score correction to recover the leverage contribution. A circular-FFT z -Cholesky attempt fails (204 vp, $\delta > 1$) due to wrap-around making V_k depend on future innovations (Section ??). The `CausalSpectralFilter` resolves this exactly, achieving 3.26 vp and delta bias -0.07 , a $63\times$ RMSE improvement available at no additional training cost.

Comparison to Markovian approximations. A standard industry approach to rough volatility is to approximate the fractional kernel by a finite sum of exponentials, yielding a *Markovian* multi-factor model that can be simulated with conventional Euler schemes. Two leading constructions are:

- **Lifted Heston** (??): approximates the fractional Riccati–Volterra equation by an n -dimensional affine jump-diffusion, with $n = 5\text{--}20$ factors needed for $H \approx 0.1$. Pricing is via Fourier inversion (characteristic function available in closed form), but the characteristic function becomes increasingly expensive as n grows.
- **3-factor Bergomi** (?): parameterises a low-dimensional Markovian system whose joint (stock, variance) dynamics approximate the rough Bergomi smile. Calibration requires non-convex optimisation over ~ 10 parameters and the approximation degrades for $H < 0.07$, where 3 factors cannot capture the roughness.

These Markovian methods share a fundamental limitation: the number of factors controls the quality of the fractional-kernel approximation, and the approximation error propagates to option prices in a way that depends on both the strike and the maturity. In contrast, the spectral surrogate *directly targets the PSD* of the fGn increments, bypassing the kernel-approximation step entirely. The implication is that the spectral surrogate’s accuracy is determined by the spectral loss convergence (Theorem ??) rather than by a factor-count parameter, and it does not degrade as $H \rightarrow 0$, a regime where Markovian approximations require increasingly many factors or break down altogether (Table ?? shows $|\Delta H| \leq 0.013$ at $H = 0.05$, where $n = 3$ Markovian approximations typically incur bias > 0.03 in recovered H).

A secondary advantage is that the spectral surrogate produces paths in a single forward pass with no matrix exponentials, no Riccati ODE integration, and no Fourier inversion, it is a pure feed-forward generator, making it trivially vectorisable and GPU-friendly. The trade-off is a one-time training cost and the W_2 convergence guarantee that holds only for the Gaussian projection of the law (Theorem ??, Assumption ??).

Connection to Paper 1. The proof structure is a one-assumption extension of ? : Bochner (stationary levels) $\rightarrow$ Cramér (stationary increments); all other lemmas carry over unchanged (see the proof-extension guide, `proof_extension_guide.tex`). The new content is the Hurst corollary (??) and the financial validation.

7 Conclusion

We have introduced the Cramér PINN, a neural-network surrogate for the rough Bergomi and rough Heston models trained to match the exact power spectral density of fractional Gaussian noise. The main contributions are: a W_2 convergence theorem extending the Bochner PINN framework to processes with stationary increments; a Hurst exponent consistency corollary; an empirical demonstration that exact PSD matching achieves 1.21–1.33 vol-point IV surface RMSE under $\rho = 0$, $1.7\times$ better than the asymptotic power-law approximation and near the Monte Carlo noise floor; a Greek bias analysis showing that the marginal V surrogate carries an irreducible ATM delta bias of -0.27 (35%) when $\rho = -0.9$ under *rough Heston*; a *CausalSpectralFilter* (causal convolution, same PSD loss) that resolves the circular-FFT wrap-around pathology,

achieving 3.26 vp vs. the BLP rBergomi reference (or **2.57 vp** with matched-PSD training, leaving only Monte-Carlo noise) and reducing the rBergomi delta bias to $\lesssim 0.01$; on rough Heston the same spectral architecture cuts the cross-model bias from +0.49 to -0.15 ($41\times$ IV RMSE improvement), and a Volterra-aware autoregressive surrogate closes the residual to MC noise (0.80 vp, bias ≤ 0.005); a Hurst estimator comparison showing that structure function achieves the smallest bias ($|\text{bias}| \leq 0.011$) for $H \in [0.05, 0.30]$ while PSD slope and R/S are unreliable below $H = 0.2$; a PSD phase diagram establishing $H \approx 0.15$ as the threshold below which exact (not asymptotic) PSD training is essential; an observation-horizon sweep recommending $T \leq 2$ yr for IV RMSE below 1.6 vp; and a live SPX Hurst estimate $\hat{H}_{\text{SPX}} = 0.145$ from the ATM IV term structure, recovered by the surrogate to within $|\Delta H| = 0.007$.

The CausalSpectralFilter introduced in Section ?? already handles $\rho \neq 0$ for rBergomi (3.26 vp vs. BLP reference). Remaining natural extensions are: (i) a joint (V, S) surrogate trained on paired paths, which would close the residual -0.07 delta bias attributable to the V-distribution gap; (ii) the deep hedging application (?), for which the causal surrogate is a drop-in path generator; and (iii) extension to other rough volatility models (RFSV, multifactor Bergomi) where the same Cramér PSD loss applies.

A Proofs

A.1 Proof of Theorem ??

We bound the three terms in (??) separately and then combine. Throughout, Assumptions ??–?? are in force.

Reduction to the Gelbrich formula. Under Assumption ??, both u_θ and u_{ref} are centred Gaussian processes; their laws are completely determined by their covariance operators C_θ, C_{ref} on $L^2([0, T])$. The Bures–Gelbrich formula (?) gives

$$W_2^2(\mathcal{L}(u_\theta), \mathcal{L}(u_{\text{ref}})) = \|C_\theta^{1/2} - C_{\text{ref}}^{1/2}\|_{\text{HS}}^2. \quad (17)$$

We bound the right-hand side in four steps.

Step 1 (Square-root Lipschitz). Because both processes have stationary increments with Cramér spectral densities satisfying Assumption ??, the covariance operators are bounded below: $C_\theta, C_{\text{ref}} \geq \mu_{\min} I$ on the spectral support $[\omega_0, \Omega]$. The operator square-root Lipschitz bound (?, Thm. X.1.1) then gives

$$\|C_\theta^{1/2} - C_{\text{ref}}^{1/2}\|_{\text{HS}} \leq \frac{1}{2\sqrt{\mu_{\min}}} \|C_\theta - C_{\text{ref}}\|_{\text{HS}}, \quad (18)$$

so that $W_2^2 \leq (4\mu_{\min})^{-1} \|C_\theta - C_{\text{ref}}\|_{\text{HS}}^2$.

Step 2 (Cramér kernel to Hilbert–Schmidt norm). By Assumption ??, the covariance kernel difference is

$$\Delta K(t, s) = \int_{\omega_0}^{\Omega} (e^{i\omega t} - 1) \overline{(e^{i\omega s} - 1)} (S_\theta(\omega) - S_{\text{ref}}(\omega)) d\omega.$$

Applying Fubini’s theorem, $|e^{i\omega t} - 1|^2 \leq 4$, and Cauchy–Schwarz on $[\omega_0, \Omega]$:

$$\|C_\theta - C_{\text{ref}}\|_{\text{HS}}^2 = \int_0^T \int_0^T |\Delta K(t, s)|^2 dt ds \leq \frac{T^2}{2\pi} \|S_\theta - S_{\text{ref}}\|_{L^2([\omega_0, \Omega])}^2. \quad (19)$$

(The factor 2π absorbs the exact integral $\int_0^1 |e^{i\omega t} - 1|^2 dt = 2 - 2\sin \omega/\omega \leq 4$.)

Combining Steps 1–2 with (??):

$$W_2(\mathcal{L}(u_\theta), \mathcal{L}(u_{\text{ref}})) \leq \frac{T}{2\sqrt{2\pi}\mu_{\min}} \|S_\theta - S_{\text{ref}}\|_{L^2([\omega_0, \Omega])} =: \frac{C_T}{\mu_{\min}^{1/2}} \|S_\theta - S_{\text{ref}}\|_{L^2([\omega_0, \Omega])}. \quad (20)$$

This is the spectral term in (??) with $C_T = T/(2\sqrt{2\pi})$.

Step 3 (Discretisation error). The spectral loss $\mathcal{L}(\theta)$ is evaluated on a discrete grid of N_s frequency points in $[\omega_0, \Omega]$. The surrogate fGn PSD S_θ is a piecewise-smooth function inheriting the Sobolev regularity $H^{H-\varepsilon}$ of fBm increments ($\beta_H = \frac{1}{2} \min(H - \frac{1}{2} + \frac{d}{2}, 1)$ for dimension d). Standard FFT quadrature theory gives

$$\|S_\theta^{\text{discrete}} - S_\theta^{\text{continuous}}\|_{L^2([\omega_0, \Omega])} \leq C_{\text{quad}} N_s^{-\beta_H},$$

with C_{quad} depending on $\|S_\theta\|_{H^{\beta_H}}$ and the geometry of the frequency grid (cf. ?, Appendix A, Lemma A.4). Inserting into (??) and setting $C_2 = C_T C_{\text{quad}}/\mu_{\min}^{1/2}$ gives the discretisation term $C_2 N_s^{-\beta_H}$.

Step 4 (Low-frequency cutoff). The spectral window $[\omega_0, \Omega]$ excludes the interval $[0, \omega_0]$ where the fBm PSD $S_H(\omega) = C_H |\omega|^{1-2H}$ diverges for $H < 1/2$. Restricting to $\omega_0 > 0$ is necessary for Assumption ?? to hold. The penalty for this restriction enters via $\mu_{\min}$: since

$$\mu_{\min} = S_H(\omega_0) = C_H \omega_0^{1-2H},$$

the constant in (??) satisfies $\mu_{\min}^{-1/2} = C_H^{-1/2} \omega_0^{-(1-2H)/2}$. For any fixed spectral gap $\|S_\theta - S_{\text{ref}}\|_{L^2}$, the W_2 bound degrades as $\omega_0^{-(1-2H)/2}$ as the window shrinks toward the origin. Setting

$$C_3 \omega_0^{-(1-2H)/2} := \frac{C_T \|S_\theta - S_{\text{ref}}\|_{L^2}}{C_H^{1/2}} \omega_0^{-(1-2H)/2}$$

identifies the third term; it vanishes as $\omega_0 \rightarrow 0$ only when $H > 1/2$ (the PSD then integrates to zero at the origin).

Combining the three terms. The triangle inequality on W_2 applied through the spectral estimate at grid resolution N_s and at frequency cutoff ω_0 gives

$$W_2(\mathcal{L}(u_\theta), \mathcal{L}(u_{\text{ref}})) \leq \underbrace{\frac{C_T}{\mu_{\min}^{1/2}} \|S_\theta - S_{\text{ref}}\|_{L^2([\omega_0, \Omega])}}_{\text{spectral term}} + \underbrace{C_2 N_s^{-\beta_H}}_{\text{discretisation}} + \underbrace{C_3 \omega_0^{-(1-2H)/2}}_{\text{cutoff}}.$$

This is (??). The three contributions are independent: the spectral term vanishes when the trained PSD matches the target; the discretisation term vanishes as the frequency grid is refined; the cutoff term reflects the irreducible cost of avoiding the $\omega = 0$ singularity. $\square$

A.2 Wood–Chan simulation

Exact simulation of fGn using circulant embedding (??).

The method exploits the fact that the covariance matrix of fGn increments $(\Delta W_1^H, \dots, \Delta W_N^H)$ is a Toeplitz matrix $\Gamma_{jk} = \frac{1}{2}(|j-k+1|^{2H} - 2|j-k|^{2H} + |j-k-1|^{2H})$. Every Toeplitz matrix can be embedded in a circulant C of size $2N$; if the embedding is positive semidefinite, the square root $C^{1/2}$ is available via FFT in $O(N \log N)$. A realisation of fGn is then $\Delta W^H = \text{Re}(\mathcal{F}^{-1}[C^{1/2}z])$ where $z \sim \mathcal{CN}(0, I)$. Davies and Harte (1987) showed the embedding is always positive semidefinite for $H \in (0, 1)$ after mirroring; ? and ? provide the practical algorithm. Our implementation is in `rbm.reference.py`, function `simulate_fBm_woodchan`.